The Maximum Nullity Bound $5d(n) \le 4(n+1)$: a Reduction to an Elliptic Curve and Kloosterman Sums

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August 25, 2026

Abstract

Let $d(n)$ be the nullity over $\mathbb{F}_2$ of the $n \times n$ bounded *Lights Out* grid. Computation suggests $5d(n) \le 4(n+1)$, with equality exactly at $n = 5 \cdot 2^k - 1$. We give a complete reduction of this inequality to a point-counting problem: writing $n + 1 = 2^k b$ with b odd, the inequality for all n is equivalent to

$$|E \cap (\mu_b \times \mu_b)| \le \tfrac{8}{5} b \qquad \text{for every odd } b,$$

where $\mu_b \subset \overline{\mathbb{F}}_2^*$ is the group of b-th roots of unity and E is the elliptic curve

$$E: \quad t + t^{-1} + s + s^{-1} = 1, \qquad \text{i.e.} \qquad t^2 s + t s^2 + ts + t + s = 0,$$

an ordinary curve over $\mathbb{F}_2$ with $\#E(\mathbb{F}_2) = 4$ and complex multiplication by $\mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$. For the two families $b = 2^r \pm 1$ we evaluate the count exactly in terms of the Kloosterman sum $K_r = \sum_{y \in \mathbb{F}_{2^r}^*} (-1)^{\mathrm{Tr}(y+y^{-1})}$, obtaining

$$G(2^r - 1) = \frac{2^r - 3 + K_r}{2}, \qquad G(2^r + 1) = \frac{2^r + 1 + K_r}{2},$$

and we deduce the bound unconditionally for those families, with $b = 5$ (i.e. $n = 5 \cdot 2^k - 1$) as the unique case of equality. A general Weil-type criterion settles all b that are large relative to $2^{\mathrm{ord}_b(2)/2}$. These regimes contain every near-extremal b, but they are a density-zero subset of all b, and we explain precisely why no soft, symplectic, Bézout-type, or coding-theoretic argument can give a constant smaller than 1, and why the outstanding cases lie in the range where no known character-sum estimate is nontrivial.

1 Introduction

Consider the game played on a simple graph $G = (V, E)$ in which every vertex carries a light, and clicking a vertex toggles that vertex and all of its neighbours. The state space is $\mathbb{F}_2^V$ and the click-to-effect map is $A + I$, where A is the adjacency matrix of G [15]. When G is the $n \times n$ grid graph $P_n \times P_n$ (no wraparound), we write

$$d(n) = \dim_{\mathbb{F}_2} \ker(A + I),$$

the *nullity* of the $n \times n$ board; this is sequence A159257 in the OEIS [10]. Exactly $2^{-d(n)}$ of all configurations are solvable, and each solvable configuration has $2^{d(n)}$ solutions [15].

Sutner [16] reduced $d(n)$ to a polynomial gcd, Hunziker, Machiavelo and Park [6] developed the Chebyshev-polynomial structure further, and the doubling law $d(2n+1) = 2d(n) + \delta_n$ conjectured in [15] was proved in [3]. Empirically, the largest nullities are attained on the families $n + 1 \in \{5, 31, 33\} \cdot 2^k$, and in every computed case

$$5d(n) \le 4(n+1), \tag{1}$$

with equality precisely when $n = 5 \cdot 2^k - 1$. We could find no statement of (1) in the literature; see Section 10. The purpose of this note is to reduce (1) to a sharply-stated arithmetic problem, to solve that problem for every family in which the bound is nearly attained, and to explain precisely which part resists all standard techniques.

1.1 Results

Throughout, b denotes an odd positive integer, $\mu_b \subset \overline{\mathbb{F}}_2^*$ is the group of b-th roots of unity, D_m is the Dickson polynomial over $\mathbb{F}_2$ determined by $D_m(t+t^{-1}) = t^m + t^{-m}$, and

$$G(m) = \deg\gcd\left(D_m(x), D_m(x+1)\right).$$

Theorem 1.1 (Reduction)**.** *For all $n \geq 1$,*

$$d(n) = G(n+1) - 2\cdot[\,3 \mid n+1\,], \qquad \text{and} \qquad G(2m) = 2G(m).$$

Consequently (1) *holds for all n if and only if $5G(b) \leq 4b$ for every odd b.*

Theorem 1.2 (Curve form)**.** *Let E be the smooth plane cubic over $\mathbb{F}_2$ with affine equation $t^2s + ts^2 + ts + t + s = 0$, equivalently $t + t^{-1} + s + s^{-1} = 1$. Then E is an elliptic curve with $\#E(\mathbb{F}_2) = 4$, its Frobenius satisfies $\pi^2 + \pi + 2 = 0$, and for every odd b*

$$N(b) := \#\left(E \cap (\mu_b \times \mu_b)\right) = 2G(b).$$

Hence (1) *is equivalent to $N(b) \leq \frac{8}{5}b$ for all odd b.*

Theorem 1.3 (Exact evaluation for $b = 2^r \pm 1$)**.** *Let $r \geq 2$, $q = 2^r$, and let $K_r = \sum_{y\in\mathbb{F}_q^*}(-1)^{\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(y+y^{-1})}$ be the binary Kloosterman sum. Then*

$$G(2^r - 1) = \frac{q - 3 + K_r}{2}, \qquad G(2^r + 1) = \frac{q + 1 + K_r}{2}.$$

In particular $G(2^r + 1) - G(2^r - 1) = 2$ for every $r \geq 2$.

Corollary 1.4. *The bound $5G(b) \leq 4b$ holds for every b of the form $2^r \pm 1$ with $r \geq 2$, and equality holds only for $b = 5$. Equivalently, $5d(n) \leq 4(n+1)$ holds whenever the odd part of $n+1$ is $2^r \pm 1$, with equality exactly for $n = 5 \cdot 2^k - 1$.*

Theorem 1.5 (General criterion)**.** *Let $b \geq 3$ be odd, $m = \mathrm{ord}_b(2)$, $q = 2^m$, and $h = (q-1)/b$. Then*

$$N(b) \leq \frac{q - 3 + 2\sqrt{q}}{h^2} + 4\sqrt{q}\left(1 - \frac{1}{h^2}\right) = 4\sqrt{q} + \frac{q - 3 - 2\sqrt{q}}{h^2}.$$

Consequently $5G(b) \leq 4b$ holds whenever

$$5\left(q - 3 - 2\sqrt{q} + 4\sqrt{q}\,h^2\right) \;\leq\; 8(q-1)h. \tag{$\star$}$$

Criterion ($\star$) holds for $h = 1$ as soon as $q \geq 8$, and in general it holds precisely when h is small compared with $\sqrt{q}$, asymptotically when $h \leq \left(\frac{2}{5} + o(1)\right)\sqrt{q}$, i.e. when $b \geq \left(\frac{5}{2} + o(1)\right)\sqrt{q}$.

Theorem 1.3 explains the extremal family completely. For $b = 2^r + 1$ the Weil bound $|K_r| \leq 2\sqrt{q}$ gives $G(b) \leq \frac{1}{2}(\sqrt{q}+1)^2$, and $\frac{5}{2}(\sqrt{q}+1)^2 \leq 4(q+1)$ is equivalent to $(3\sqrt{q}-1)(\sqrt{q}-3) \geq 0$, i.e. to $r \geq 4$. At $r = 2$, that is $b = 5$, the Weil bound fails by exactly $\frac{1}{2}$: it gives $G(5) \leq 4.5$ while the truth is $G(5) = 4 = \frac{4}{5}\cdot 5$. The gap is closed by the congruence $K_r \equiv -1 \pmod 4$ [7], which forces $K_2 \in \{-1, 3\}$ and hence $G(5) \leq 4$. *The unique equality case of* (1) *is therefore the unique place where the Weil bound for a Kloosterman sum is not quite sharp enough on its own.*

Section 8 explains why the remaining cases are hard: the symplectic, Bézout and coding-theoretic arguments all give exactly $d(n) \leq n$ and no better (a bound attained at $n = 4$), while character sums are nontrivial only when b is large compared with $2^{\mathrm{ord}_b(2)/2}$, which fails for almost every b. The saving grace, and the reason the present results are not vacuous, is that the b for which $G(b)/b$ is large are exactly the b for which 2 has small multiplicative order, i.e. exactly the ones the character-sum methods do reach.

All numerical claims below are verified by `code/upper_bound_check.py`, which shares no code with the rest of this repository.

2 Row chasing, the transfer matrix, and Sutner's gcd

Definition 2.1. Let $u_0 = 0$, $u_1 = 1$ and $u_{k+1} = xu_k + u_{k-1}$ in $\mathbb{F}_2[x]$, and put $f_k = u_{k+1}$, so that

$$f_0 = 1, \qquad f_1 = x, \qquad f_{k+1} = xf_k + f_{k-1}.$$

These are the Fibonacci (or Chebyshev-like) polynomials of [16, 6, 4]; $\deg f_n = n$ and $f_n = \sum_j \binom{n-j}{j} x^{n-2j}$ over $\mathbb{F}_2$.

Let P denote the adjacency matrix of the path P_n and let $M = I + P$. A configuration $y \in \mathbb{F}_2^{n\times n}$ lies in $\ker(A + I)$ if and only if, writing $r_i \in \mathbb{F}_2^n$ for its i-th row and setting $r_0 = r_{n+1} = 0$,

$$r_{i-1} + Mr_i + r_{i+1} = 0 \qquad (1 \le i \le n). \tag{2}$$

Lemma 2.2. *$d(n) = \dim\ker f_n(M)$.*

Proof. Rewriting (2) as $r_{i+1} = Mr_i + r_{i-1}$ and inducting gives $r_i = f_{i-1}(M)r_1$ for $1 \le i \le n+1$. Hence $y \mapsto r_1$ is an injection of $\ker(A + I)$ into $\ker f_n(M)$, and conversely every $r_1 \in \ker f_n(M)$ generates a configuration satisfying (2) together with both boundary conditions. ■

Proposition 2.3 (Symplectic form of the recursion)**.** *Let $T = \begin{pmatrix} 0 & I \\ I & M \end{pmatrix}$ act on $\mathbb{F}_2^{2n}$ by $(r_{i-1}, r_i) \mapsto (r_i, r_{i+1})$, and let $J = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$. Then $T^\intercal JT = J$, so $T \in \mathrm{Sp}(2n, \mathbb{F}_2)$. The subspaces $L = 0 \oplus \mathbb{F}_2^n$ and $L' = \mathbb{F}_2^n \oplus 0$ are Lagrangian and*

$$d(n) = \dim\left(L \cap T^{-n}L'\right) \le n.$$

Proof. $JT = \begin{pmatrix} I & M \\ 0 & I \end{pmatrix}$ and $T^\intercal(JT) = \begin{pmatrix} 0 & I \\ I & M + M \end{pmatrix} = J$ using $M^\intercal = M$ and $\mathrm{char}\,\mathbb{F}_2 = 2$. The form is alternating because $\langle (a, b), (a, b)\rangle = 2a^\intercal b = 0$. Both L and L' are isotropic of dimension n. By Lemma 2.2 a kernel element is exactly a starting vector $(0, r_1) \in L$ whose n-th iterate (r_n, r_{n+1}) lies in L'. ■

Lemma 2.4 (Sutner [16])**.** *$d(n) = \deg\gcd\left(f_n(x),\ f_n(x + 1)\right)$.*

Proof. P is a Jacobi matrix (tridiagonal with nonzero off-diagonal entries), hence nonderogatory, and its characteristic polynomial satisfies the same recursion as f, so it equals f_n. Therefore $\mathbb{F}_2^n \cong \mathbb{F}_2[x]/(f_n)$ as a module over $\mathbb{F}_2[P]$, with P acting as multiplication by x. Under this identification $M = I + P$ acts as multiplication by $x + 1$, so $f_n(M)$ acts as multiplication by $f_n(x + 1)$. In $R = \mathbb{F}_2[x]/(f)$, the kernel of multiplication by g is the ideal generated by $f/\gcd(f, g)$, of dimension $\deg\gcd(f, g)$. Apply Lemma 2.2. ■

3 Dickson polynomials and the 2-adic reduction

Definition 3.1. Let $D_m(x) = x\,u_m(x) = xf_{m-1}(x) \in \mathbb{F}_2[x]$ for $m \ge 1$, and $D_0 = 0$.

Lemma 3.2. *D_m is the Dickson polynomial of degree m with parameter 1 [8]: it satisfies $D_0 = 0$, $D_1 = x$, $D_{m+1} = xD_m + D_{m-1}$, and*

$$D_m\big(t + t^{-1}\big) = t^m + t^{-m}.$$

Consequently $D_{2m} = D_m^2$.

Proof. The recursion is immediate from that of u. The sequence $g_m(t) = t^m + t^{-m}$ satisfies $g_0 = 0$, $g_1 = t + t^{-1}$ and $(t + t^{-1})g_m = g_{m+1} + g_{m-1}$, i.e. the same recursion in $x = t + t^{-1}$; the identity follows by induction. Since $t^{2m} + t^{-2m} = (t^m + t^{-m})^2$ and $t + t^{-1}$ takes infinitely many values in $\overline{\mathbb{F}}_2$, we get $D_{2m} = D_m^2$. ∎

Definition 3.3. For odd b set

$$R_b = \left\{t + t^{-1} : t \in \mu_b \setminus \{1\}\right\}, \qquad W(b) = R_b \cap (R_b + 1).$$

Since t and t^{-1} give the same value and $t = t^{-1}$ only for $t = 1$, we have $|R_b| = \frac{b-1}{2}$, and $0 \notin R_b$.

Lemma 3.4. *For odd b we have $u_b = \tau_b^2$ where $\tau_b = \prod_{\alpha \in R_b}(x + \alpha)$ is squarefree of degree $\frac{b-1}{2}$, and $D_b = x\tau_b^2$. Moreover $\tau_b(0) = 1$, and $\tau_b(1) = 0$ if and only if $3 \mid b$.*

Proof. By Lemma 3.2, $D_b(\alpha) = 0$ with $\alpha = t + t^{-1}$ if and only if $t^{2b} = 1$, i.e. $t^b = 1$; so the roots of D_b are $\{0\} \cup R_b$. As $\deg D_b = b = 1 + 2\,|R_b|$ and $x \mid D_b$ exactly once (because $0 \notin R_b$), each $\alpha \in R_b$ is a double root. $0 \notin R_b$ gives $\tau_b(0) \neq 0$, and $1 \in R_b$ means $t + t^{-1} = 1$ for some $t \in \mu_b \setminus \{1\}$, i.e. $t^2 + t + 1 = 0$, i.e. t is a primitive cube root of unity, which lies in μ_b exactly when $3 \mid b$. ∎

Proposition 3.5. *For odd b,*

$$G(b) = 2\,|W(b)| + 2 \cdot [\,3 \mid b\,].$$

Moreover $|W(b)|$ is even, so $4 \mid G(b)$ whenever $3 \nmid b$.

Proof. Write $\widetilde{\tau}(x) = \tau_b(x + 1)$, so $D_b(x) = x\tau_b^2$ and $D_b(x + 1) = (x + 1)\widetilde{\tau}^2$. The roots of τ_b are R_b and those of $\widetilde{\tau}$ are $R_b + 1$, both simple, so $\deg\gcd(\tau_b, \widetilde{\tau}) = |W(b)|$. For an irreducible $p \notin \{x, x + 1\}$, $v_p(\gcd(D_b(x), D_b(x + 1))) = 2v_p(\gcd(\tau_b, \widetilde{\tau}))$. For $p = x$: $v_x(D_b) = 1$ by Lemma 3.4 and $v_x(D_b(x + 1)) = 2v_x(\widetilde{\tau}) = 2[\,3 \mid b\,]$, so the minimum is $[\,3 \mid b\,]$; the case $p = x + 1$ is symmetric. Adding the contributions gives the formula. Finally, if $\alpha \in W(b)$ then $\alpha, \alpha + 1 \in R_b$, hence $\alpha + 1 \in W(b)$ as well; the involution $\alpha \mapsto \alpha + 1$ is fixed-point free in characteristic 2, so $|W(b)|$ is even. ∎

Proof of Theorem 1.1. The doubling law is immediate from $D_{2m} = D_m^2$:

$$G(2m) = \deg\gcd\big(D_m(x)^2, D_m(x + 1)^2\big) = 2G(m).$$

For the first identity write $n + 1 = 2^k b$ with b odd. By Lemmas 3.2 and 3.4,

$$xf_n(x) = D_{n+1}(x) = D_b(x)^{2^k} = \big(x\tau_b^2\big)^{2^k}, \qquad \text{so} \qquad f_n(x) = x^{2^k-1}\tau_b^{2^{k+1}}.$$

Now compute $d(n) = \deg\gcd(f_n(x), f_n(x + 1))$ valuation by valuation, writing $\widetilde{\tau}(x) = \tau_b(x + 1)$, so that $f_n(x + 1) = (x + 1)^{2^k-1}\widetilde{\tau}^{\,2^{k+1}}$. Both τ_b and $\widetilde{\tau}$ are squarefree, and by Lemma 3.4 we have $\tau_b(0) \neq 0$, $\widetilde{\tau}(1) = \tau_b(0) \neq 0$, and $\widetilde{\tau}(0) = \tau_b(1) = 0$ exactly when $3 \mid b$. Hence

$$\begin{aligned} v_x(\gcd) &= \min\big(2^k - 1,\ 2^{k+1}[\,3 \mid b\,]\big) = \big(2^k - 1\big)\,[\,3 \mid b\,], \\ v_{x+1}(\gcd) &= \min\big(2^{k+1}[\,3 \mid b\,],\ 2^k - 1\big) = \big(2^k - 1\big)\,[\,3 \mid b\,], \\ \sum_{p \notin \{x, x+1\}} v_p(\gcd)\deg p &= 2^{k+1}\deg\gcd(\tau_b, \widetilde{\tau}) = 2^{k+1}\,|W(b)|\,. \end{aligned}$$

Therefore, using Proposition 3.5 and the doubling law,

$$d(n) = 2^{k+1}\,|W(b)| + 2\left(2^k - 1\right)[\,3 \mid b\,], \qquad G(n+1) = 2^k G(b) = 2^{k+1}\,|W(b)| + 2^{k+1}[\,3 \mid b\,],$$

whence $G(n+1) - d(n) = 2 \cdot [\,3 \mid b\,] = 2 \cdot [\,3 \mid n+1\,]$. For the final statement: if $5G(b) \le 4b$ for all odd b, then $5d(n) \le 5G(n+1) = 5 \cdot 2^k G(b) \le 4 \cdot 2^k b = 4(n+1)$. Conversely, suppose $5d(n) \le 4(n+1)$ for all n and let b be odd. If $3 \nmid b$, take $n = b-1$, so $d(n) = G(b)$ and $5G(b) \le 4b$. If $3 \mid b$, take $n = 2^k b - 1$, so $5\left(2^k G(b) - 2\right) \le 4 \cdot 2^k b$; dividing by 2^k and letting $k \to \infty$ gives $5G(b) \le 4b$. ■

Remark 3.6. The formula $d(n) = 2^{k+1}\,|W(b)| + 2(2^k - 1)[\,3 \mid b\,]$ recovers, in one line, the doubling behaviour of [6, 3]: all the dependence on k is explicit, and all the arithmetic content is in the single quantity $|W(b)|$.

4 The elliptic curve

Definition 4.1. Let $E \subset \mathbb{P}^2_{\mathbb{F}_2}$ be the projective closure of

$$t^2 s + t s^2 + ts + t + s = 0,$$

that is, $T^2 S + T S^2 + TSU + TU^2 + SU^2 = 0$.

Multiplying $t + t^{-1} + s + s^{-1} = 1$ by ts gives exactly this equation, so on the open set $ts \ne 0$ the curve is the locus $t + t^{-1} + s + s^{-1} = 1$.

Lemma 4.2. *E is a smooth plane cubic, hence an elliptic curve of genus* 1*. Its three points at infinity are* $(1:0:0)$, $(0:1:0)$, $(1:1:0)$*, its only affine $\mathbb{F}_2$-point is* $(0,0)$*, so $\#E(\mathbb{F}_2) = 4$ and the Frobenius π satisfies $\pi^2 + \pi + 2 = 0$. In particular E is ordinary with complex multiplication by $\mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$, and for $q = 2^m$,*

$$\#\left\{(t,s) \in (\mathbb{F}_q^*)^2 : t + t^{-1} + s + s^{-1} = 1\right\} = q - 3 - a_m,$$

where $a_0 = 2$, $a_1 = -1$ and $a_m = -a_{m-1} - 2a_{m-2}$.

Proof. The partial derivatives of $F = T^2S + TS^2 + TSU + TU^2 + SU^2$ in characteristic 2 are $F_T = S^2 + SU + U^2$, $F_S = T^2 + TU + U^2$, $F_U = TS$. A singular point needs $TS = 0$; if $T = 0$ then $F_S = U^2 = 0$ forces $U = 0$ and then $F_T = S^2 = 0$, so $S = 0$, impossible in $\mathbb{P}^2$; the case $S = 0$ is symmetric. Setting $U = 0$ gives $TS(T+S) = 0$, the three listed points. Among the four affine $\mathbb{F}_2$-points only $(0,0)$ satisfies the equation, so $\#E(\mathbb{F}_2) = 4 = 2 + 1 - a$ with $a = -1$, and $\pi^2 - a\pi + 2 = 0$ [14]. Since a is odd, E is ordinary, and $\mathbb{Z}[\pi] = \mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$. The affine count over $\mathbb{F}_q$ removes the three points at infinity and the point $(0,0)$ (note $t = 0$ forces $s = 0$ on E), and $\#E(\mathbb{F}_q) = q + 1 - a_m$ with $a_m = \pi^m + \bar{\pi}^m$, whence the recursion. ■

Proof of Theorem 1.2. Only $N(b) = 2G(b)$ remains. Let $\alpha \in W(b)$. Then $\alpha = t + t^{-1}$ and $\alpha + 1 = s + s^{-1}$ for some $t, s \in \mu_b \setminus \{1\}$, and (t,s) satisfies the equation of E; each such α arises from exactly $2 \cdot 2 = 4$ pairs, namely $t^{\pm 1}$ and $s^{\pm 1}$. The remaining points of $E \cap (\mu_b \times \mu_b)$ are those with $t = 1$ or $s = 1$: $t = 1$ forces $s + s^{-1} = 1$, which has two solutions $s \in \mu_3$ when $3 \mid b$ and none otherwise, and symmetrically for $s = 1$. Hence $N(b) = 4\,|W(b)| + 4[3 \mid b\,] = 2G(b)$ by Proposition 3.5. ■

Remark 4.3. Theorem 1.2 converts (1) into an *unlikely intersection* statement: the curve E meets the "torsion grid" $\mu_b \times \mu_b$, a set of b^2 points, in at most $\frac{8}{5}b$ points. Bézout gives only $N(b) \le 2b$ (the equation is quadratic in s for fixed t), which is exactly the trivial bound $d(n) \le n$ again.

5 Exact evaluation for $b = 2^r \pm 1$

Throughout this section $q = 2^r$ and $\mathrm{Tr} = \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}$.

Lemma 5.1. *(a) If $b = q-1$, so that $\mu_b = \mathbb{F}_q^*$, then $R_b = \{\alpha \in \mathbb{F}_q^* : \mathrm{Tr}(\alpha^{-1}) = 0\}$.*

(b) If $b = q+1$, so that $\mu_b \subset \mathbb{F}_{q^2}^*$ is the norm-one subgroup, then $R_b = \{\alpha \in \mathbb{F}_q^* : \mathrm{Tr}(\alpha^{-1}) = 1\}$.

Proof. In both cases $\alpha = t + t^{-1}$ means $t^2 + \alpha t + 1 = 0$ with $\alpha \neq 0$; substituting $t = \alpha z$ turns this into $z^2 + z = \alpha^{-2}$, which is solvable in $\mathbb{F}_q$ if and only if $\mathrm{Tr}(\alpha^{-2}) = 0$, i.e. $\mathrm{Tr}(\alpha^{-1}) = 0$ (the trace is invariant under squaring). In case (a), t ranges over $\mathbb{F}_q^*$ and $\alpha \neq 0$ excludes $t = 1$. In case (b), $t \in \mu_{q+1}$ means $t^{q+1} = 1$, i.e. $t\bar{t} = 1$ for the nontrivial automorphism of $\mathbb{F}_{q^2}/\mathbb{F}_q$; then $t^{-1} = \bar{t}$, so $\alpha = t + \bar{t} \in \mathbb{F}_q$, and $t \notin \mathbb{F}_q$ (equivalently $t \neq 1$, since $\mu_{q+1} \cap \mathbb{F}_q^* = \{1\}$) precisely when $z^2 + z = \alpha^{-2}$ has *no* root in $\mathbb{F}_q$, i.e. $\mathrm{Tr}(\alpha^{-1}) = 1$. ■

Lemma 5.2. *Let $S_1 = \sum_{\alpha\neq 0,1}(-1)^{\mathrm{Tr}(\alpha^{-1})}$, $S_2 = \sum_{\alpha\neq 0,1}(-1)^{\mathrm{Tr}((\alpha+1)^{-1})}$ and $S_3 = \sum_{\alpha\neq 0,1}(-1)^{\mathrm{Tr}(\alpha^{-1}+(\alpha+1)^{-1})}$, the sums being over $\mathbb{F}_q \setminus \{0,1\}$. Then*

$$S_1 = S_2 = -1 - (-1)^r, \qquad S_3 = -1 + K_r.$$

Proof. $\sum_{\alpha\in\mathbb{F}_q^*}(-1)^{\mathrm{Tr}(\alpha^{-1})} = \sum_{\beta\in\mathbb{F}_q^*}(-1)^{\mathrm{Tr}\,\beta} = -1$, and removing $\alpha = 1$ subtracts $(-1)^{\mathrm{Tr}(1)} = (-1)^r$; this gives S_1, and $\alpha \mapsto \alpha+1$ permutes $\mathbb{F}_q \setminus \{0,1\}$, giving $S_2 = S_1$. For S_3, note $\alpha^{-1} + (\alpha+1)^{-1} = (\alpha^2+\alpha)^{-1}$. As α runs over $\mathbb{F}_q \setminus \{0,1\}$, $y = \alpha^2 + \alpha$ runs twice over $\{y \neq 0 : \mathrm{Tr}(y) = 0\}$, so with $[\mathrm{Tr}(y) = 0] = \frac{1}{2}\left(1 + (-1)^{\mathrm{Tr}\,y}\right)$,

$$S_3 = 2\sum_{\substack{y\neq 0\\ \mathrm{Tr}(y)=0}}(-1)^{\mathrm{Tr}(y^{-1})} = \sum_{y\neq 0}(-1)^{\mathrm{Tr}(y^{-1})} + \sum_{y\neq 0}(-1)^{\mathrm{Tr}(y+y^{-1})} = -1 + K_r. \qquad ■$$

Proof of Theorem 1.3. Let $b = q-1$. By Lemma 5.1(a),

$$|W(b)| = \sum_{\alpha\neq 0,1}\frac{1+(-1)^{\mathrm{Tr}(\alpha^{-1})}}{2}\cdot\frac{1+(-1)^{\mathrm{Tr}((\alpha+1)^{-1})}}{2} = \frac{(q-2)+S_1+S_2+S_3}{4} = \frac{q-5-2(-1)^r+K_r}{4}.$$

Since $3 \mid 2^r - 1$ exactly when r is even, i.e. exactly when $(-1)^r = 1$, Proposition 3.5 gives in both parities

$$G(2^r-1) = 2\,|W| + 2[\,3 \mid b\,] = \frac{q-3+K_r}{2}.$$

For $b = q+1$, Lemma 5.1(b) replaces each factor $\frac{1+(-1)^{\mathrm{Tr}}}{2}$ by $\frac{1-(-1)^{\mathrm{Tr}}}{2}$, so

$$|W(b)| = \frac{(q-2)-S_1-S_2+S_3}{4} = \frac{q-1+2(-1)^r+K_r}{4},$$

and $3 \mid 2^r + 1$ exactly when r is odd, so again both parities give

$$G(2^r+1) = \frac{q+1+K_r}{2}. \qquad ■$$

Proof of Corollary 1.4. Weil's bound gives $|K_r| \leq 2\sqrt{q}$, and by [7] $K_r \equiv -1 \pmod 4$.

Case $b = 2^r+1$. We must show $5(q+1+K_r) \leq 8(q+1)$, i.e. $5K_r \leq 3(q+1)$. If $2\sqrt{q}\cdot 5 \leq 3(q+1)$ we are done; this is $3q - 10\sqrt{q} + 3 = (3\sqrt{q}-1)(\sqrt{q}-3) \geq 0$, which holds for $q \geq 9$, i.e. $r \geq 4$. For $r = 3$, $K_3 = -5 < 0 \leq \frac{3\cdot 9}{5}$. For $r = 2$ we have $q = 4$ and $|K_2| \leq 4$ with $K_2 \equiv -1 \pmod 4$, so $K_2 \in \{-1, 3\}$ and $5K_2 \leq 15 = 3(q+1)$, with equality if and only if $K_2 = 3$. Indeed $K_2 = 3$, giving $G(5) = 4$ and $5G(5) = 20 = 4 \cdot 5$.

Case $b = 2^r - 1$. We must show $5(q-3+K_r) \le 8(q-1)$, i.e. $5K_r \le 3q+7$. From $|K_r| \le 2\sqrt{q}$ it suffices that $3q - 10\sqrt{q} + 7 = (3\sqrt{q}-7)(\sqrt{q}-1) \ge 0$, which holds for $\sqrt{q} \ge 7/3$, i.e. $r \ge 3$. For $r=2$, $b=3$ and $K_2 = 3$ give $G(3) = 2$ and $10 \le 12$.

Uniqueness of equality. Equality for $b = 2^r+1$ forces $K_r = \frac{3(q+1)}{5}$, and $\frac{3(q+1)}{5} > 2\sqrt{q}$ for $q \ge 16$; the only remaining possibilities are $r=2$ (which does give equality) and $r=3$ (where $\frac{3\cdot 9}{5} \notin \mathbb{Z}$). Equality for $b = 2^r - 1$ forces $K_r = \frac{3q+7}{5}$, which exceeds $2\sqrt{q}$ for $q \ge 16$ and is not an integer for $q \in \{4, 8\}$. Finally, by Theorem 1.1, $5d(n) = 4(n+1)$ with $n+1 = 2^k b$ requires $5G(b) = 4b$ and $3 \nmid b$, i.e. $b = 5$. ■

Remark 5.3. Theorem 1.3 yields $G(2^r+1) - G(2^r-1) = 2$ for all $r \ge 2$, hence, after applying the correction terms of Theorem 1.1,

$$d(2^r) - d(2^r-2) = \begin{cases} 4 & r \text{ even}, \\ 0 & r \text{ odd}. \end{cases}$$

This is the bounded-grid analogue of the theorem of Goshima and Yamagishi [5] that the corresponding *torus* nullity satisfies $d(2^r+1) = d(2^r-1)+4$, which they also prove using an elliptic curve.

Corollary 5.4. *For all $k \ge 0$,*

$$d\big(5\cdot 2^k - 1\big) = 2^{k+2}, \qquad d\big(31\cdot 2^k - 1\big) = 5\cdot 2^{k+2}, \qquad d\big(33\cdot 2^k - 1\big) = 11\cdot 2^{k+1} - 2.$$

Proof. By Theorem 1.1, $d(2^k b - 1) = 2^{k+1}\,|W(b)| + 2(2^k-1)[3 \mid b]$, and $|W(b)| = \frac{1}{2}\,(G(b) - 2[3 \mid b])$ by Proposition 3.5. Theorem 1.3 with $r=2$ gives $G(5) = \frac{4+1+K_2}{2} = 4$ since $K_2 = 3$, and with $r=5$ and $K_5 = 11$ gives $G(31) = \frac{32-3+11}{2} = 20$ and $G(33) = \frac{32+1+11}{2} = 22$. Hence $|W(5)| = 2$, $|W(31)| = 10$ and $|W(33)| = 10$, and the three formulas follow. ■

Remark 5.5. The last two formulas were originally suggested by computation, in the notation $g(b,k) = b\cdot 2^{k-1} - 1$: $d(g(31,k)) = 5\cdot 2^{k+1}$ and $d(g(33,k)) = 11\cdot 2^k - 2$. They are now theorems. The same argument evaluates d on the family $b\cdot 2^k - 1$ for *any* fixed odd b, the only input being the single integer $G(b)$; the content of Theorem 1.3 is that $G(b)$ is a Kloosterman sum when $b = 2^r \pm 1$.

6 A general character-sum criterion

Lemma 6.1. *On E,* $\operatorname{div}(t) = 2(0,0) - (1:0:0) - (1:1:0)$ *and* $\operatorname{div}(s) = 2(0,0) - (0:1:0) - (1:1:0)$. *Consequently, for integers a, c the function $t^a s^c$ is a D-th power in $\overline{\mathbb{F}}_q(E)^*$ only if $D \mid a$ and $D \mid c$.*

Proof. The only zero of t is $(0,0)$ and its only poles are the points at infinity with $T \ne 0$, namely $(1:0:0)$ and $(1:1:0)$; the multiplicities are forced by $\deg\operatorname{div}(t) = 0$ together with simplicity of the poles at the two smooth infinite points. The claim for s is symmetric. Then $\operatorname{div}(t^a s^c)$ has coefficient $-a$ at $(1:0:0)$ and $-c$ at $(0:1:0)$, and a D-th power has divisor divisible by D. ■

Proof of Theorem 1.5. Let $m = \operatorname{ord}_b(2)$, $q = 2^m$, so $\mu_b \subset \mathbb{F}_q^*$ is the subgroup of index $h = (q-1)/b$. Let H be the group of multiplicative characters of $\mathbb{F}_q^*$ trivial on μ_b, so $|H| = h$ and $\mathbf{1}_{\mu_b} = \frac{1}{h}\sum_{\chi \in H}\chi$. Let $U = E \setminus \{(0,0), (1:0:0), (0:1:0), (1:1:0)\}$, so that $U(\mathbb{F}_q) = \{(t,s) \in (\mathbb{F}_q^*)^2 : t + t^{-1} + s + s^{-1} = 1\}$. Then

$$N(b) = \sum_{P \in U(\mathbb{F}_q)} \mathbf{1}_{\mu_b}(t(P))\,\mathbf{1}_{\mu_b}(s(P)) = \frac{1}{h^2}\sum_{\chi,\psi \in H} S(\chi,\psi), \qquad S(\chi,\psi) = \sum_{P \in U(\mathbb{F}_q)} \chi(t(P))\psi(s(P)).$$

The principal term is $S(\mathbf{1},\mathbf{1}) = \#U(\mathbb{F}_q) = q - 3 - a_m \le q - 3 + 2\sqrt{q}$ by Lemma 4.2 and the Hasse bound. For $(\chi,\psi) \ne (\mathbf{1},\mathbf{1})$, fix a character λ of $\mathbb{F}_q^*$ of order $q-1$ and write $\chi = \lambda^a$, $\psi = \lambda^c$ with

$(a, c) \not\equiv (0,0) \bmod q-1$; then $\chi(t)\psi(s) = \lambda(t^a s^c)$ and, by Lemma 6.1, $t^a s^c$ is not a $(q-1)$-st power in $\overline{\mathbb{F}}_q(E)^*$. Weil's bound for multiplicative character sums along a curve [13, 1] therefore gives

$$|S(\chi, \psi)| \leq (2g - 2 + \#\operatorname{supp}\operatorname{div}(t^a s^c))\sqrt{q} \leq (0+4)\sqrt{q},$$

since $g = 1$ and the support is contained in the four removed points. Summing the $h^2 - 1$ nonprincipal terms gives the stated inequality. Multiplying $N(b) \leq 4\sqrt{q} + (q - 3 - 2\sqrt{q})h^{-2}$ by $5h^2$ and using $\frac{8}{5}bh^2 = \frac{8}{5}(q-1)h$ turns the requirement $N(b) \leq \frac{8}{5}b$ into $(\star)$. For $h = 1$, $(\star)$ reads $5(q+2\sqrt{q}-3) \leq 8(q-1)$, i.e. $10\sqrt{q} \leq 3q+7$, which holds for $q \geq 8$. In general the left side of $(\star)$ grows like $20\sqrt{q}\,h^2$ and the right side like $8qh$, so $(\star)$ holds exactly in the range $h \lesssim \frac{2}{5}\sqrt{q}$. ■

Remark 6.2. A two-dimensional discrete Fourier calibration over $\mathbb{F}_{2^m}$ for $m \leq 11$ shows $\max_{(\chi,\psi)\neq(\mathbf{1},\mathbf{1})} |S(\chi, \psi)| \leq 3.73\sqrt{q}$, consistent with the constant 4 above.

7 What is proved unconditionally

Theorem 7.1. *Write $n + 1 = 2^k b$ with b odd. Then $5d(n) \leq 4(n + 1)$ holds whenever*

(i) $b = 2^r \pm 1$ for some $r \geq 2$ (Corollary 1.4), or

(ii) b satisfies criterion $(\star)$ of Theorem 1.5, or

(iii) $b \leq 10^5$ (verified by computation, Section 9).

In these cases equality holds if and only if $b = 5$, i.e. $n = 5 \cdot 2^k - 1$.

The two unconditional regimes (i) and (ii) are thin: among the 5999 odd b with $3 \leq b \leq 12000$, exactly 24 are covered by (i) and a further 37 by (ii), so 61 in all, or 1.02%. However, they contain *every* b for which $G(b)/b$ is anywhere near 4/5: the largest value of $G(b)/b$ over all $b \leq 12000$ *not* covered by (i) or (ii) is 0.2222, attained at $b = 99$, versus the target 0.8. This is not a coincidence, and it is the main structural finding of this note:

> $G(b)/b$ is large exactly when 2 has small multiplicative order modulo b, and that is exactly the regime in which character sums over μ_b are nontrivial.

8 Why standard bounds do not suffice

8.1 Every soft argument stops at $d(n) \leq n$

Proposition 2.3 exhibits $d(n)$ as the dimension of an intersection of two Lagrangian subspaces of a symplectic $\mathbb{F}_2$-space of dimension $2n$; such an intersection can have any dimension between 0 and n, and no property of the specific T beyond symplecticity is used. Bézout applied to $E \cap (\mu_b \times \mu_b)$ gives $N(b) \leq 2b$, i.e. the same bound. Restriction to the first row (Lemma 2.2) gives an injection of the kernel into $\mathbb{F}_2^n$, again the same bound. And the bound $d(n) \leq n$ *is attained*, at $n = 4$, where $d(4) = 4$: an $\mathbb{F}_2$-computation shows $n = 4$ is the only $n < 4000$ with $d(n) = n$. Hence no argument that is insensitive to the arithmetic of b can produce any constant $c < 1$ in $d(n) \leq c\,n$.

8.2 Coding bounds are vacuous

The kernel is a binary linear code $\mathcal{K} \subseteq \mathbb{F}_2^{n^2}$ of dimension $d(n)$, and by Lemma 2.2 it is equivalent to a code of length n and the same dimension. The Singleton, Griesmer, and Plotkin bounds all require a lower bound on the minimum distance of that length-n code, and no nontrivial such bound is known here; applied to the ambient length n^2 they are vacuous by an enormous margin. The one genuinely code-theoretic input we do use — the congruence $K_r \equiv -1 \pmod 4$ — comes from coding theory [7], and it is exactly what settles the extremal case $b = 5$.

8.3 The character-sum barrier

Theorem 1.5 is nontrivial only when $b \gtrsim \sqrt{q} = 2^{\mathrm{ord}_b(2)/2}$. For a typical odd b, $\mathrm{ord}_b(2)$ is of size $b^{1-o(1)}$, so $q = 2^{\mathrm{ord}_b(2)}$ is exponentially larger than b and the bound $4\sqrt{q}$ dwarfs the target $\frac{8}{5}b$; the estimate says nothing at all. This is the same obstruction that makes multiplicative-subgroup sums hard in general: the strongest available results, of Bourgain–Glibichuk–Konyagin type [2], give a power saving for subgroups of size $b > q^{\varepsilon}$ with $\varepsilon > 0$ fixed, whereas here b can be as small as $q^{o(1)}$ (indeed $b \approx \log_2 q$ in the worst case). No known method gives a nontrivial estimate for $\sum_{t\in\mu_b} \chi(\cdot)$ in that range, and any proof of (1) in full must, by Theorem 1.2, do exactly that or avoid character sums entirely.

8.4 Unlikely intersections do not help in characteristic p

Over $\mathbb{C}$, the Manin–Mumford / Ihara–Serre–Tate theorem bounds $\left|C \cap \mu_\infty^2\right|$ for a curve C that is not a torsion coset, uniformly in the torsion. In characteristic 2, *every* point of $\mathbb{G}_m^2(\overline{\mathbb{F}}_2)$ is torsion, so $E \cap \mu_\infty^2 = E(\overline{\mathbb{F}}_2)$ is infinite. The correct characteristic-p statement, due to Moosa and Scanlon [9], describes such intersections as finite unions of translates of "F-sets". This is a qualitative structure theorem; it yields no uniform inequality of the form $|E \cap (\mu_b \times \mu_b)| \le c\,b$ with $c < 2$, which is precisely what (1) asks for.

8.5 The bound is a genuinely arithmetic statement

Any proof of (1) must separate the values

$$\frac{G(5)}{5} = 0.8, \qquad \frac{G(3)}{3} = \frac{G(33)}{33} = \frac{2}{3}, \qquad \frac{G(31)}{31} = 0.6452, \qquad \frac{G(2^r \pm 1)}{2^r \pm 1} \to \frac{1}{2}.$$

By Theorem 1.3 the last is a Kloosterman-sum fluctuation of size $O(q^{-1/2})$ around $\frac{1}{2}$. A uniform argument giving $G(b) \le \frac{4}{5}b$ would in particular have to be tight at $b = 5$ and lossy by only 17% at $b = 33$, while all these numbers are values of character sums. We regard this as strong evidence that (1) is not accessible by any "soft" method, and that a complete proof needs an estimate for $N(b)$ that is valid uniformly for subgroups $\mu_b \subset \mathbb{F}_q^*$ of arbitrarily small relative size — a statement of independent interest well beyond the present state of the art.

9 Computational evidence

The script `code/upper_bound_check.py` verifies, independently of the rest of this repository:

(1) f_n from the recursion agrees with $\sum_j \binom{n-j}{j} x^{n-2j}$ for $n \le 200$;

(2) $\deg \gcd(f_n(x), f_n(x+1))$ agrees with the nullity of the $n^2 \times n^2$ Lights Out matrix computed by Gaussian elimination, for $n \leq 12$;

(3) $d(n) = G(n+1) - 2[\,3 \mid n+1\,]$ and $G(2m) = 2G(m)$ for $n, m \leq 3000$;

(4) $\#U(\mathbb{F}_{2^m}) = 2^m - 3 - a_m$ for $m \leq 13$, and $N(b) = 2G(b)$ for all odd $b \leq 200$ with $\mathrm{ord}_b(2) \leq 13$;

(5) the formulas of Theorem 1.3 and the congruence $K_r \equiv -1 \pmod 4$, for $2 \leq r \leq 13$;

(6) $5G(b) \leq 4b$ for all odd $b \leq 10^5$, the only equality being $b = 5$, and the largest ratio $G(b)/b$ apart from $b = 5$ being $\frac{22}{33} = \frac{2}{3}$;

(7) the coverage statistics quoted in Section 7.

b	$G(b)$	$G(b)/b$	b	$G(b)$	$G(b)/b$
5	4	0.8000	2049	1058	0.5163
33	22	0.6667	2047	1056	0.5159
3	2	0.6667	4097	2072	0.5057
31	20	0.6452	4095	2070	0.5055
257	144	0.5603	513	254	0.4951
255	142	0.5569	511	252	0.4932

Table 1: The twelve largest values of $G(b)/b$ for odd $b \leq 12000$. Every entry lies in a family $2^r \pm 1$ or is 3 or $33 = 2^5 + 1$; all are covered by Corollary 1.4 or Theorem 1.5.

Correspondingly, the only $n < 4000$ with $5d(n) = 4(n+1)$ are

$$n \in \{4, 9, 19, 39, 79, 159, 319, 639, 1279, 2559\} = \{5 \cdot 2^k - 1\},$$

and the largest value of $\frac{5d(n)}{4(n+1)}$ over $n < 3000$ outside that family is 0.8321, attained at $n = 2111 = 33 \cdot 2^6 - 1$. For $b = 5$ everything is explicit: $f_{5\cdot 2^k - 1} = x^{2^k - 1}(x^2 + x + 1)^{2^{k+1}}$, the gcd is $(x^2 + x + 1)^{2^{k+1}}$, and $d(5 \cdot 2^k - 1) = 2^{k+2}$.

10 Relation to the literature

Sutner [15, 16] introduced $d(n)$ and proved Lemma 2.4; Sarkar [11] and Sarkar–Barua [12] study the same quantity through π-polynomials and rule out totally irreversible σ-grids in certain congruence classes; Hunziker, Machiavelo and Park [6] prove the identity $F_{n2^k}(x) = x^{2^k - 1} F_n(x)^{2^k}$ underlying Theorem 1.1; Goldwasser, Klostermeyer and Ware [4] treat the equivalent parity-domination problem; and [3] proves Sutner's doubling conjecture. Goshima and Yamagishi [5] prove the *torus* analogues $d(2n) = 2d(n)$ and $d(2^r + 1) = d(2^r - 1) + 4$ "using an elliptic curve", and Yamagishi [17, 18, 19] develops the curve-theoretic approach for the Trisentis game and for σ-automata on square grids.

Remark 10.1 (Attribution)**.** We were unable to obtain the text of [18], whose title ("Elliptic curves over finite fields and reversibility of additive cellular automata on square grids") makes it very likely that the curve E of Section 4, or a curve isogenous to it, already appears there, and possibly also some form of Theorem 1.3. Priority for the curve and for point-count formulas of that type should be assumed to rest with Yamagishi until [18] can be checked. What we have not found anywhere in the literature is the inequality (1) itself, in any form, nor the reduction of it to a subgroup point count, nor the identification of $b = 5$ as the unique equality case via the Weil/congruence gap for K_2.

11 Conclusion

The inequality $5d(n) \le 4(n+1)$ is equivalent to the single arithmetic statement $|E \cap (\mu_b \times \mu_b)| \le \frac{8}{5}b$ for all odd b, for one explicit ordinary elliptic curve $E/\mathbb{F}_2$ with CM by $\mathbb{Z}[\frac{1+\sqrt{-7}}{2}]$. This statement is *proved* for the families $b = 2^r \pm 1$, where it is nearly tight, and for all b that are large relative to $2^{\mathrm{ord}_b(2)/2}$; and it is verified for all $b \le 10^5$. Its extremal case $b = 5$ is exactly the case in which the Weil bound for a binary Kloosterman sum is off by $\frac{1}{2}$ and must be repaired by the congruence $K_r \equiv -1 \pmod 4$. What remains open is a uniform point count for a curve intersected with a multiplicative subgroup of arbitrarily small relative size — a problem for which, as of today, no technique is known that beats the trivial bound.

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